

Chapter 10: Amazing Air

Key Terms and Definitions

Atmosphere: The thick blanket-like layer of air surrounding the Earth, extending up to approximately 320 kilometres above the surface.

Air (as a Mixture): Air is a mixture with variable composition. It has no definite chemical formula and no fixed boiling point. Each constituent of air retains its individual properties.

Active and Inactive Components: Oxygen (21%) serves as the **active component** because it supports life and combustion, while nitrogen (78%) acts as the **inactive component** and dilutes the activity of oxygen.

Respiration: A chemical process occurring in living cells in which oxygen reacts with glucose to produce energy, carbon dioxide, and water.

Inhaled Air and Exhaled Air: **Inhaled air** is the air taken into the body during breathing. **Exhaled air** is the air released from the lungs and contains a higher concentration of carbon dioxide.

Stomata: Minute openings present on the surface of plant leaves through which plants exchange gases.

Photosynthesis: The process by which green plant parts use carbon dioxide and water in the presence of chlorophyll and sunlight to synthesize glucose (stored as starch) and release oxygen.

Combustion: A process in which a substance reacts with oxygen to release energy in the form of heat and light. It is commonly called burning.

Fuel: Materials such as wood, coal, kerosene, petrol, diesel, LPG, and CNG that undergo combustion to release heat and light energy.

Environmental Pollutants & Their Impacts

Carbon Dioxide (CO₂): Excess accumulation of carbon dioxide in the atmosphere increases Earth's surface temperature, contributing to global warming.

Sulfur Dioxide (SO₂): A pungent gas that can contribute to acid rain, increasing the acidity of soil and water, and can contribute to smog, which reduces visibility.